

ENGR 102 Summer Bridge

Day 4 Instructor Key

Comparisons, Exact Boundaries, and Ordered Branches

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Evaluate exact-boundary comparisons.

5 points

```
low = 10
high = 15
value = 15
within = low <= value <= high
above = value > high
print(within, above)
```

Evaluate every comparison and explain which operator controls the upper boundary.

Answer and explanation

10 <= 15 and 15 <= 15 are both True, so within is True. The strict comparison 15 > 15 is False. Exact output: True False. The <= operator includes equality at high.

Question 2. Trace the first matching ordered branch.

5 points

```
reading = 25

if reading > 25:
    status = "high"
elif reading >= 20:
    status = "normal"
else:
    status = "low"

print(status)
```

Name the first matching branch, give exact output, and state what happens at 25 and 25.1.

Answer and explanation

At 25, reading > 25 is False and reading >= 20 is True, so the elif branch assigns normal. At 25.1, the first branch assigns high. Exact supplied output: normal.

Question 3. Program construction**10 points**

Write one complete Python program that satisfies every requirement.

- Read temperature as a float.
- Print invalid when temperature is below -40 or above 80.
- For valid input, classify cold below 15, warm from 15 through 30 inclusive, and hot above 30.
- Print exactly Status: followed by the classification using one if/elif/elif/else chain.

Answer and explanation

```
temperature = float(input())

if temperature < -40 or temperature > 80:
    print("invalid")
elif temperature < 15:
    print("Status: cold")
elif temperature <= 30:
    print("Status: warm")
else:
    print("Status: hot")
```

The invalid branch protects the announced domain first. Once input is valid, the earlier cold branch removes values below 15, so temperature <= 30 correctly represents the remaining inclusive warm interval.

Boundary evidence: -40 is cold, 15 and 30 are warm, 30.1 is hot, and 80.1 is invalid.

Scoring guidance

2 input, 2 invalid domain, 3 ordered boundaries, 2 mutually exclusive structure/exact output, 1 boundary evidence.